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(54) Title: **Wild track ai: real-time animal detection and mobile notification system.**

(51) International Classification: A01K 29/00, G08B 21/02, A01M 29/16, G08B 25/01 H04W 4/02 (21) Application No: 202621023514 (31) Priority Document No: -- (32) Priority Date: -- (86) International Application No: -- Filing Date: -- (87) International Publication No: -- (61) Patent of Addition to Application No: -- Filing Date: -- (62) Divisional to Application No: -- Filing Date: --	(71) Name of Applicant(s): 1. Prof. Mayuri Mangesh Gawade, 2. Hariom Papansinh Thakur, 3. Kartik Dilip Thorat, 4. Prashant Jaywant Thorat, 5. Om Ashok Vhargal, (72) Name of Inventor(s): 1. Prof. Mayuri Mangesh Gawade 2. Hariom Papansinh Thakur 3. Kartik Dilip Thorat 4. Prashant Jaywant Thorat 5. Om Ashok Vhargal
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(57) Abstract:

The present invention, titled "WildTrack AI: Real-Time Animal Detection and Mobile Notification System," relates to an intelligent wildlife monitoring system configured to detect and report animal presence in forest-border and sensitive rural areas. The system comprises an image acquisition module including a high-resolution surveillance camera with night-vision capability for continuous monitoring. An edge processing module, implemented using an embedded computing device with AI acceleration, executes a trained deep learning-based object detection model to identify animal species in real time using predefined confidence threshold filtering. Upon validated detection, a communication module utilizing GSM, Wi-Fi, or LTE connectivity transmits structured detection data to a cloud server. The cloud infrastructure stores detection logs including captured image, timestamp, GPS location, species label, and confidence score, and manages secure user authentication. A mobile application generates instant push notifications to registered users with image preview, mapped location details, and alert severity indication. The invention enables timely preventive action, reduces human-wildlife conflict, and enhances wildlife monitoring efficiency through real-time automated detection, edge computing, and cloud-integrated mobile alert mechanisms.



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[PART II—SEC.

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THE GAZETTE OF INDIA : EXTRAORDINARY

[PART II—SEC. 3(ii)]

FORM 2

THE PATENT ACT 1970
(39 of 1970)The Patents Rules, 2003
PROVISIONAL COMPLETE SPECIFICATION
(See section 10 and rule 13)

1. TITLE OF THE INVENTION : WildTrack AI: Real-Time Animal Detection and Mobile Notification System.

2. APPLICANT (S) (a) (a)

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3. PREAMBLE TO THE DESCRIPTION

COMPLETE

1. The following specification describes the invention titled "WildTrack AI: Real-Time Animal Detection and Mobile Notification System."
2. The present invention relates generally to the field of wildlife monitoring, artificial intelligence-based surveillance, and real-time alert systems. More particularly, the invention relates to an integrated system and method for automated detection of wildlife presence in forest-border and sensitive rural areas using deep learning techniques, edge computing, and cloud-based mobile notification services.
3. Human-wildlife conflict has become a significant concern in regions adjacent to forest zones, agricultural lands, and rural settlements. Conventional monitoring systems rely heavily on manual patrolling, basic motion sensors, or delayed camera trap analysis, which are often inefficient, slow, and unable to provide real-time alerts. These limitations result in delayed response, crop damage, livestock loss, and potential risks to human life
4. The proposed WildTrack AI system addresses these challenges by providing an intelligent, automated, and real-time wildlife detection and alert mechanism. The system comprises:
 - An Image Acquisition Unit, consisting of a surveillance camera with night vision capability installed in forest-border or sensitive zones, configured to continuously capture live video streams.
 - An Edge Processing Unit, implemented using an embedded computing device such as Raspberry Pi or NVIDIA Jetson, configured to execute a trained deep learning-based object detection model (for example, YOLO architecture) to process video frames in real time and identify animal species based on predefined confidence threshold values.
 - A Communication Module, including Wi-Fi, GSM, or LTE connectivity, configured to transmit processed detection data to a remote cloud infrastructure.
 - A Cloud Server and Database, configured to store detection records including captured image, timestamp, GPS coordinates, detected species label, and confidence score, and further configured to manage user authentication and notification services.
 - A Mobile Application, configured to send instant push notifications to registered users, including forest officials and local residents, with image preview and precise location details whenever an animal is detected.

4. DESCRIPTION (Description shall start from next page.)

(a) Image Acquisition Module

The Image Acquisition Module is configured to capture real-time visual data from monitored locations.

- **High-Resolution Camera:**
The system employs a high-resolution surveillance camera capable of capturing clear image frames and video streams to ensure accurate object detection and classification.
- **Night-Vision / Infrared Support:**
The camera includes infrared (IR) or low-light night-vision capability, enabling continuous monitoring during nighttime or low-visibility environmental conditions.
- **Motion-Trigger Mechanism:**
The module may incorporate a motion detection sensor or frame-differencing algorithm configured to activate recording and processing only when motion is detected, thereby optimizing computational resources and power consumption.

(b) Processing Module

The Processing Module is responsible for real-time analysis of captured image frames.

- **Edge Device with GPU / AI Accelerator:**
The module comprises an embedded edge computing device such as Raspberry Pi, NVIDIA Jetson, or any equivalent AI-enabled processor, configured to perform on-device inference without reliance on continuous cloud processing.
- **Deep Learning Model (YOLO-Based Object Detection):**
A trained deep learning object detection model based on the YOLO (You Only Look Once) architecture is deployed on the edge device to identify and classify animal species within captured frames.
- **Confidence Threshold Filtering:**
The system applies predefined confidence threshold values to filter detections, ensuring that only predictions exceeding a minimum confidence level are considered valid, thereby reducing false positives and unnecessary alerts.

(c) Communication Module

The Communication Module enables transmission of validated detection data to a remote server.

- **GSM / Wi-Fi / LTE Connectivity:**
The module supports multiple communication technologies including GSM, Wi-Fi, and LTE networks to ensure reliable data transmission in rural and forest-border environments.
- **MQTT / HTTP-Based Data Transmission:**
Detection data including image snapshots, species label, timestamp, GPS coordinates, and confidence score are transmitted securely using lightweight communication protocols such as MQTT or HTTP.

(d) Cloud & Database Module

The Cloud and Database Module manages centralized storage, processing, and user management functionalities.

- **Firebase / Cloud Server:**
A cloud-based backend infrastructure, such as Firebase or equivalent cloud server architecture, is configured to receive and process transmitted detection data.
- **Storage of Detection Logs:**

The system stores structured detection records including captured image, detected species, date and time, GPS location, and associated confidence score for future reference and analytics.

- **User Authentication System:**

A secure authentication mechanism is implemented to manage registered users, including forest officials and local residents, ensuring authorized access to detection alerts and stored data.

(e) Mobile Application Module

The Mobile Application Module provides end-user interaction and real-time alert functionality.

- **Real-Time Push Notifications:**

The system generates and delivers instant push notifications to registered users whenever an animal detection event exceeds the predefined confidence threshold.

- **Image Preview of Detected Animal:**

Each notification includes a preview image of the detected animal to enable quick visual verification.

- **Location Mapping:**

The application integrates GPS-based mapping functionality to display the precise detection location on a digital map interface.

- **Alert Severity Indication:**

The system may categorize alerts based on predefined risk levels (for example, high-risk species vs. low-risk species), enabling users to prioritize response actions accordingly.

7. ABSTRACT OF THE INVENTION (to be given along with complete specification on separate page) :

The present invention, titled "**WildTrack AI: Real-Time Animal Detection and Mobile Notification System**," relates to an intelligent wildlife monitoring system configured to detect and report animal presence in forest-border and sensitive rural areas. The system comprises an image acquisition module including a high-resolution surveillance camera with night-vision capability for continuous monitoring. An edge processing module, implemented using an embedded computing device with AI acceleration, executes a trained deep learning-based object detection model to identify animal species in real time using predefined confidence threshold filtering.

Upon validated detection, a communication module utilizing GSM, Wi-Fi, or LTE connectivity transmits structured detection data to a cloud server. The cloud infrastructure stores detection logs including captured image, timestamp, GPS location, species label, and confidence score, and manages secure user authentication. A mobile application generates instant push notifications to registered users with image preview, mapped location details, and alert severity indication.

The invention enables timely preventive action, reduces human-wildlife conflict, and enhances wildlife monitoring efficiency through real-time automated detection, edge computing, and cloud-integrated mobile alert mechanisms.

Note: -

*Repeat boxes in case of more than one entry.

be signed by the applicant(s) or by authorized registered patent agent.

*Name of the applicant should be given in full family name in the beginning .

*Complete address of the applicant should be given stating the postal index state and country.

*Strike out the column which is/are not applicable

5. CLAIMS (not applicable for provisional specification, Claims should start with the preamble — "%/we claim" on separate page)

Claim 1 (Independent Claim) :A mobile-based wildlife detection and alert system comprising:

- a smartphone application configured to capture real-time images using an onboard camera;
- a remote inference server configured to receive captured images through a network interface;
- a deep learning-based object detection engine deployed on the server and trained to detect predefined wildlife species;
- a species risk classification module configured to categorize detected wildlife into dangerous and non-dangerous classes;
- a global positioning system (GPS) tagging module configured to associate geographic coordinates with each detection event; and
- a notification module configured to generate an alert when a detected species is classified as dangerous.

Claim 2 (Dependent Claim) :The system of Claim 1, wherein the deep learning-based object detection engine comprises a YOLO based neural network architecture trained on a wildlife-specific dataset.

Claim 3 (Dependent Claim) : The system of Claim 1, wherein the notification module transmits real-time alerts to the detecting user through in-application notifications or short message service (SMS).

Claim 4 (Independent Claim – System Architecture) A wildlife monitoring architecture comprising:

- a mobile client device configured to capture and transmit image data;
- a RESTful application programming interface (API) configured to process incoming image requests;
- a server-side inference engine configured to return detection results including bounding box coordinates, class labels, and confidence scores;
- and a database configured to store detection history along with timestamp and geographic metadata. This claim protects your architecture specifically.

Claim 5 (Dependent Claim) :The system of Claim 4, wherein stored detection history is retrievable and visualizable on a map interface within the mobile application. **Claim 6 (Independent Claim – Risk-Based Alert Logic)** A computer-implemented method for wildlife risk notification comprising:

- receiving an image captured from a mobile device;
- detecting one or more wildlife objects using a trained deep learning model;
- assigning a predefined risk level to each detected wildlife species;
- and automatically triggering a warning notification when the assigned risk level exceeds a predefined safety threshold. This protects your dangerous vs non-dangerous classification logic.

Claim 7 (Dependent Claim): The method of Claim 6, wherein the warning notification includes the detected species name and associated geographic coordinates.

Claim 8 (Dependent Claim – Multi-User System) :The system of Claim 1 further comprising a user authentication module configured to uniquely associate **detection events with registered users**.

Claim 9 (Dependent Claim – Rural Deployment): The system of Claim 1, wherein the inference engine is configured to operate on a lightweight server environment suitable for low-bandwidth rural deployment.

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6.DATE AND SIGNATURE (to be given at the end of last page of specification)

Guide Signature	Student SIGNATURE			
Pro. Mauri Gavde	Prashant Jaywant Thorat	Om Ashok Vhargal	KARTIK DILIP Thorat	Hariomsinh Papansinh Thakur
				